

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – COMPARITIVE ANALYSIS

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO4	Assessment:	AT1 - COMPARITIVE ANALYSIS
Weightage:	30 % of CIA	Total Marks:	100
CO4 Statement:	<i>Identify the components and structure of mobile application for Android and Windows OS-based devices.</i>		
Student Name:	KAVIYA M	Reg. No.:	192421050
Section / Batch:	SLOT D - 2024	Date:	18.08.26

Code of Conduct

I KAVIYA M 192421050 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- **Duration: 60 minutes**
- **Number of questions : 01**

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

15.

A security researcher is evaluating two mobile operating systems for resistance against malicious applications. Compare Android and BlackBerry OS based on application permissions, sandboxing, process isolation, and security architecture. Analyze their strengths and weaknesses and conclude which provides stronger protection.

COMPARATIVE ANALYSIS OF ANDROID AND BLACKBERRY OS SECURITY

1. Introduction

Mobile operating systems need strong security mechanisms to protect users from malicious applications and unauthorized access. **Android** and **BlackBerry OS** use different approaches to protect applications, system resources, and user data. This report compares both operating systems based on **application permissions, sandboxing, process isolation, and overall security architecture.**



2. Objective

The objectives of this comparative analysis are:

- To study the security mechanisms used by Android and BlackBerry OS.
- To compare their application permission systems.
- To analyze their sandboxing and process isolation techniques.

- To identify the strengths and weaknesses of both operating systems.
- To determine which operating system provides stronger protection against malicious applications.

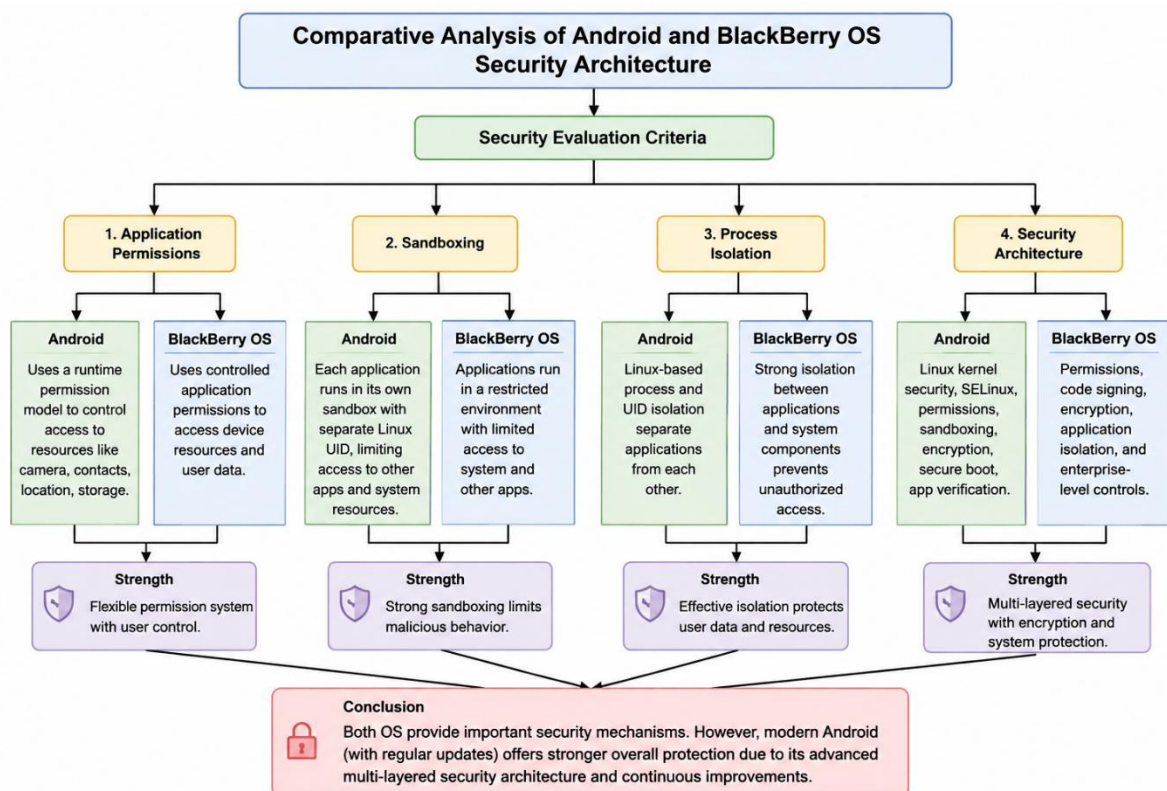
3. Security Architecture

Android

Android is based on the Linux kernel and uses multiple security layers. Each application normally receives its own Linux user identity and runs in an isolated environment. Android also uses application permissions, sandboxing, SELinux, encryption, secure boot, and application verification mechanisms.

BlackBerry OS

BlackBerry OS was designed with a strong focus on security, particularly for enterprise users. It provides application permissions, application isolation, code signing, encryption, and controlled access to system resources. Its security architecture was designed to prevent unauthorized applications from accessing sensitive information.



4. Comparative Analysis

Security Aspect	Android	BlackBerry OS
Application Permissions	Uses permissions to control access to camera, contacts, location, storage, etc. Modern Android allows users to grant permissions at runtime.	Uses application permissions to control access to device resources and user information.
Sandboxing	Each application runs in its own sandbox and normally cannot directly access another application's private data.	Applications operate in controlled environments with restrictions on access to system resources.
Process Isolation	Linux user IDs and process isolation separate applications from each other.	Provides strong separation between applications and protected system components.
Security Architecture	Uses Linux security, SELinux, sandboxing, permissions, encryption, verified boot, and application security mechanisms.	Uses permissions, code signing, encryption, application isolation, and enterprise security controls.
Malware Protection	Uses application verification and multiple layers of security to detect and restrict harmful applications.	Strong application controls and signing mechanisms reduce the risk of unauthorized applications.
Flexibility	Highly flexible and supports a large application ecosystem.	More controlled and security-focused, particularly for enterprise environments.
Major Advantage	Strong modern multi-layer security architecture.	Strong security-by-design approach and enterprise protection.
Major Limitation	Large application ecosystem creates a larger attack surface.	Legacy BlackBerry OS platforms have limited modern security updates and application support.

5. Strengths and Weaknesses

Android

Strengths:

- Strong application sandboxing.
- Separate user identity for applications.
- Fine-grained permission system.
- SELinux provides additional protection.
- Encryption protects stored data.
- Verified boot helps protect the operating system.
- Large ecosystem with continuous security improvements.

Weaknesses:

- Large number of applications increases the attack surface.
- Malicious applications may attempt to misuse granted permissions.
- Security updates can vary between manufacturers and devices.
- Older Android versions may contain unpatched vulnerabilities.

BlackBerry OS

Strengths:

- Security was a major design priority.
- Strong application isolation.
- Controlled application permissions.
- Code signing helps prevent unauthorized software.
- Strong encryption capabilities.
- Well suited for enterprise environments.

Weaknesses:

- Older BlackBerry OS platforms have limited current security support.
- Smaller application ecosystem.
- Less flexibility compared with Android.

- Legacy platforms do not benefit from many modern mobile security improvements.

6. Analysis

From the comparison, both operating systems provide important protection against malicious applications. **BlackBerry OS traditionally followed a more controlled security model**, making it particularly suitable for organizations that required strict security and centralized management.

Android uses a layered security model. Application sandboxing prevents applications from freely accessing each other's data, while permissions control access to sensitive resources. Additional mechanisms such as SELinux, encryption, verified boot, and application verification provide multiple levels of protection.

Therefore, Android provides greater flexibility but also has a larger attack surface because of its extensive application ecosystem.

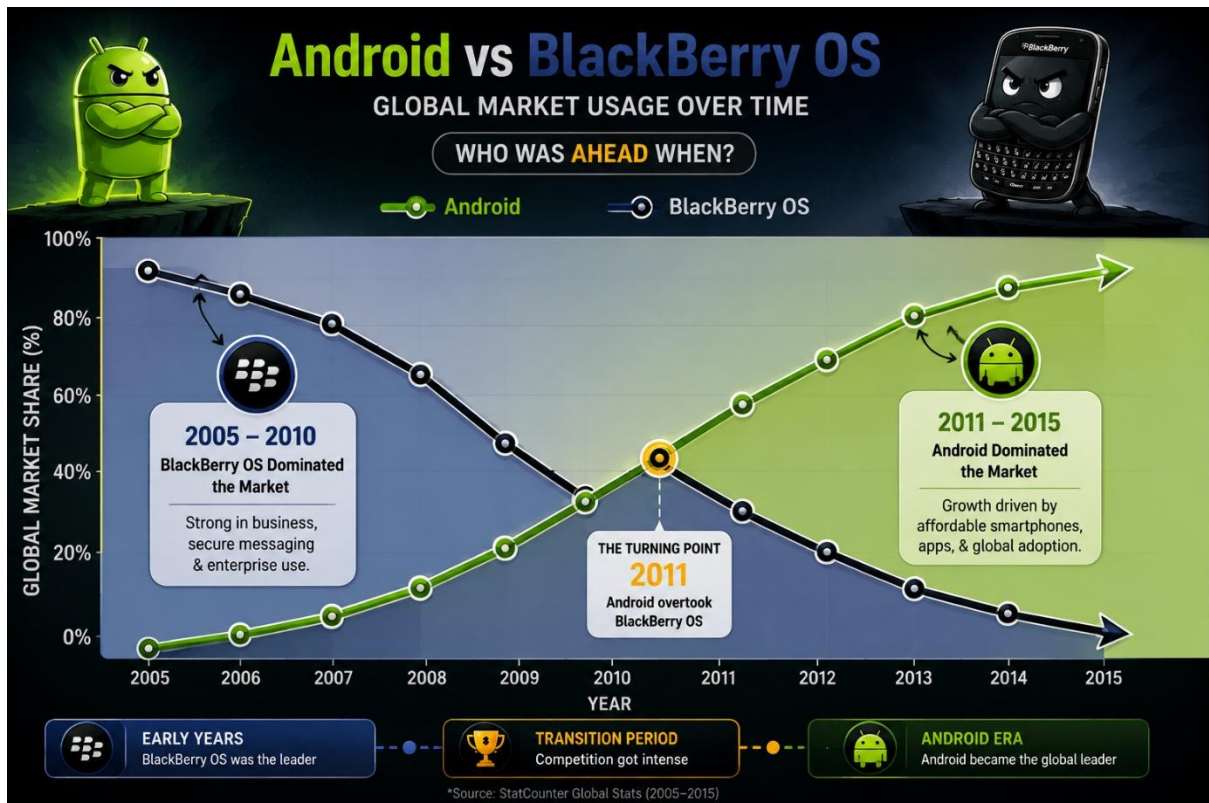




7. Conclusion

Both Android and BlackBerry OS provide mechanisms such as **application permissions, sandboxing, and process isolation** to protect mobile devices from malicious applications.

Historically, **BlackBerry OS provided stronger security through its highly controlled and security-focused architecture**, especially for enterprise environments. However, when considering **modern operating systems**, an up-to-date Android device provides stronger overall protection because of its multi-layered security architecture, including **sandboxing, SELinux, encryption, verified boot, permissions, and continuous security updates**.



Final conclusion: Modern Android provides stronger overall protection against malicious applications, provided that the device is kept updated and applications are installed from trusted sources.